# COLLISION RECORDS, ARITHMETIC, AND RAW LOCAL INVERSION IN A TRIANGULAR LORENTZ GAS

QIAN QI

Abstract. We investigate the unchanged exact-label raw pointwise local limit problem for actual returns in a triangular Lorentz gas. On the original positive paired endpoint source, we prove a finite-count persistence inequality that pays every physical sheet at a previously fixed positive total residence, regardless of flatness, critical endpoint structure, physical comparison reach or stationarity. The corresponding essential-height constant is independent of the fixed positive residence cutoff. We also compare actual stationary critical arms at an adaptive, smaller physical reach and pay their complete positive outgoing-trace shortage by the original uncut optical volume. Combining the two mechanisms with the inherited critical-arm comparison reduces the original pointwise error to the original unmatched shrinking-residence positive source. We formulate an explicit untruncated residence-moment sufficient criterion for its decay without asserting that the Lorentz source satisfies that missing criterion. The original pointwise theorem, section, exact occupation and displacement labels, arithmetic zero classes and scalar-first conditional-law order are unchanged. We further prove a two-ended positive-envelope comparison on nonstationary physical sheets. All losses of either endpoint trace are paid by the original volume on a common roof window; only unallocated comparison capacity remains in the positive remainder. This comparison requires neither a stationary incoming roof nor a positive lower bound for the paired trace.

We construct the complete admissible comparison pool at each original roof, with a Borel allocation and physical overlap one. The resulting accepted source is maximal within the fixed-window envelope-certificate class and pointwise dominates every compatible fixed-family allocation at the same total cost. This removes angular block selection from the microscopic reduction. Collision-uniform decay of the remaining canonical capacity shortage is distinguished from finite-count exhaustion.

## 1. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write $T_R$ for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; $p$ is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and $\tau_+$ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let $Y_R$ be the union of the four open coordinate squares of half-side $h_0$ centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and $q_R$ is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

*Date*: October 14, 2026. A2-DYN, revision 108.

2020 *Mathematics Subject Classification.* 37D50, 37A50, 60F05, 28A75.

*Key words and phrases.* Lorentz gas, exact collision records, positive paired traces, stationary endpoint roofs, physical residence, angular comparison, critical entrances.

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At $m$ flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{1.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual $n$th return has collision count $m$ and roof $L$. Denote its density by $p_{n,R}(k, m, t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let $\Delta_j$ be the least clearance of flight $j$ from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d'_j = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{1.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{1.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

1.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R, \vartheta)$ and $\mathsf{P}_j(R, \vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let $\chi_j$ be their frequency partition. At the real maximum $a_j$ of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and $\ell$ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{1.4}$$

Only the charts active at $R$ are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)| \longrightarrow 0. \tag{1.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on $R$.

1.2. **Comparison at a stationary endpoint.** The previous endpoint-distance flows compare nearby trajectories of the same exact event. Their paired coarea densities are $c^\varepsilon/|a'|$ and $c^\varepsilon/|b'|$, where $c^\varepsilon$ is the positive minimum of the weighted traces at the two ends of a maximal regular segment. These formulas remove the relative-position degeneracy but still require nonzero endpoint speed along a useful comparison path.

The argument below uses intervals of actual angular trajectories instead of a single comparison path. If all trajectories in a comparison interval contain a one-sided roof translate of every source trajectory, the integral of their positive trace controls the source at its original roof. The ratio of the two integrated traces and the physical overlap are the only numerical costs. Stationary endpoint velocity and rapid local variation of the minimum do not enter that ratio. Theorem 4.2 verifies it on critical arms with an explicit level-width condition and the original uncut optical variation. Theorem 4.3 also pays the entire comparison-trace shortage. Its whole-family height bound includes all relative positions, without a lower original residence cutoff.

The pointwise problem is still (1.5). The new theorems estimate specified positive source families at that topology, not a replacement finite-norm problem. Theorem 5.1 retains a nonnegative remainder with a precise comparison-trace deficit. Its vanishing on the complete physical family remains the last requirement in the resulting scalar reduction.

1.3. **Comparison without a stationary entrance.** The two endpoint traces play symmetric roles in the next comparison. Their positive maximum supplies comparison capacity, while their absolute difference is controlled by the original optical variation. Theorem 10.4 proves the resulting essential-height estimate on actual physical sheets with moving entrances and exits. Lemma 10.3 localizes the entire comparison segment near the target roof by interval inclusion alone. It replaces the common-incoming-roof argument, rather than assuming its conclusion. Theorem 11.1 then retains, with its original weight, precisely the unallocated part of the microscopic source. No physical coverage or microscopic height limit is inferred from finite-count definability or from a model calculation.

## 2. The inherited physical source in the required form

Fix the radius, collision count and exact displacement and occupation labels. On the disjoint original space of regular pairing sheets, write $a(\theta) < u < b(\theta)$ for a maximal roof segment. On the regular set away from $g = 0$, put

$$q = (\alpha_0, \alpha_m), \quad F = L, \quad g = \nabla_q F = R(-p_0, p_m), \quad z = |g|, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i.$$

Here $\omega = |\partial_{p_0}\alpha_m|^{-1}$ and $\chi = |\det D_q(F, \arg g)|$. The physical angular density is

$$A(u,\theta) = \frac{\omega(u,\theta)}{4\pi c_* \chi(u,\theta)}, \qquad f^\varepsilon = AW^\varepsilon, \qquad c^\varepsilon(\theta) = \min\{f^\varepsilon(a+,\theta), f^\varepsilon(b-,\theta)\}.$$

The initial section visit is counted and the terminal section visit is tested separately, as in (1.1). Every angular sheet carries this same exact event; its incident physical component is part of its identity.

**Input statement 2.1** (Positive pairing and local concentration)**.** The original paired measure is $c^\varepsilon(\theta)\, d\theta\, du$ on $a < u < b$. For $0 < \Delta$ let $\mathsf{T}^\varepsilon_\Delta$ be its roof density restricted to $0 < \ell = b - a < \Delta$, without an additional selector. At every previously fixed $r, \Delta, \varepsilon > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r \mathsf{T}^\varepsilon_\Delta\|_\infty \le C\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}, \qquad A_r f(t) = \frac{1}{2r}\int_{t-r}^{t+r} f(u)\, du. \tag{2.1}$$

The leading constant is independent of the previously fixed widths. For fixed $m, R, n, k$, all nonnegative angular restrictions use the same coarea kernels outside a roof-null set. Countable selections use common versions. The uniform order is $\sup_R \operatorname{ess\,sup}_t$; one universal null set for all radii is not asserted.

This is the positive barycentric source and complete paired-average estimate of Supplement Theorem 4.2 and equation (5.1). The paired-average estimate follows there from the original weighted source and the actual uncut inverse-gradient volume source. It is not an assumed essential-height estimate for the paired source. The same inherited analytic input gives the actual uncut source

$$v^0(t)\,dt = F_{\#}(I_m z^{-1}\mathbf{1}_\Pi \nu_R^*), \qquad \limsup_m \sup_{R,n,k} m^2 \|A_\rho v^0\|_\infty \le C \tag{2.2}$$

for every previously fixed $\rho > 0$, with a width-independent leading constant. It is finite at fixed count. On each regular segment,

$$|\partial_u f^\varepsilon(u,\theta)| \le C(1+\varepsilon^{-1})A(u,\theta)I_m(u,\theta)/z(u,\theta). \tag{2.3}$$

These are the actual uncut volume and weighted optical variation statements in Supplement Input statement 3.2 and Input statement 3.1. The latter records the maximal-segment map and one-sided trace comparison; the underlying statements are Supplement Theorem 7.1 and Theorem 6.1. The uncut bound is Supplement Theorem 11.3. No unweighted pointwise height bound for $v^0$ is assumed. The bound is used only on a previously fixed positive window after the physical sum.

The Borel physical sheet summation retains multiplicity one; its historical proof and both endpoint trace conventions are unchanged.

**Input statement 2.2** (Regular critical entrance and scalar reduction)**.** At a regular nongrazing backward-critical extension $q_*$, the incoming roof is constant and

$$f^\varepsilon(a+,\theta) = \frac{N_*}{e_\perp(\theta)^{\mathsf T} K_* e_\perp(\theta)}, \quad N_* = \frac{\omega(q_*)W^\varepsilon(q_*)}{4\pi c_*}, \quad \lambda_\circ I \le K_* \le \Lambda_\circ I, \tag{2.4}$$

where $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. The incoming trace can be positive. A singular stationary extension is not included in this formula merely because its incoming roof is critical.

Under the original outer schedule

$$\varepsilon(B) = A_0 B^{-1/12}, \qquad q_0^* = 1/8, \qquad r_0 = c_0\varepsilon(B)^4, \qquad \delta = 6q_0^* r_0, \tag{2.5}$$

the preceding positive remainder $\mathsf{U}_B^{\mathrm{rem}}$ is the original short paired source on failure of the relative clock and both endpoint-distance clocks. With

$$\mathcal{H}_M(T) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} T(t),$$

one has

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B^{\mathrm{rem}}) + C_M B^{-1/384}. \tag{2.6}$$

Along this same schedule, the original pointwise assertion is equivalent to $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{rem}}) \to 0$.

The trace formula is Supplement Proposition 7.2; the scalar statement is Supplement Theorem 8.2. The earlier relative and endpoint-distance comparisons remain available and unchanged. The new comparison is applied specifically to their positive failure source below.

## 3. Positive comparison without an endpoint velocity

We work on the original disjoint space of physical pairing sheets. An angular point on that space is denoted by $\xi$; its measure is $d\theta$, its endpoint roofs are $a(\xi) < b(\xi)$, and

$$c(\xi) = \min\{f^{\varepsilon}(a+,\xi), f^{\varepsilon}(b-,\xi)\}, \qquad \ell(\xi) = b(\xi) - a(\xi).$$

The suppressed superscript on $c$ is always the original $\varepsilon$. In these coordinates the positive paired measure is $c(\xi)\,d\theta\,du$ on $a(\xi) < u < b(\xi)$. In particular

$$\mathsf{T}^{\varepsilon}_{\Delta}(t) = \int c(\xi)\mathbf{1}_{\{0<\ell(\xi)<\Delta\}}\mathbf{1}_{\{a(\xi)<t<b(\xi)\}}\,d\theta. \tag{3.1}$$

All sheets carry the same fixed exact event. No occupation class is pooled, and coincident physical faces retain their incident component.

Fix $0 < \delta < \Delta$ and $r > 0$. A comparison pair consists of Borel angular sets $I_j, J_j$ on an actual physical sheet, and a sign $\tau_j \in \{-1, 1\}$. Its source has $0 < \ell \le \delta$ on $I_j$; its comparison set has $0 < \ell < \Delta$ on $J_j$. For almost every $(\xi, \zeta) \in I_j \times J_j$ assume the corresponding inclusion

$$\begin{aligned} \tau_j = 1: &\qquad a(\zeta) \le a(\xi), \quad b(\zeta) \ge b(\xi) + r, \\ \tau_j = -1: &\qquad a(\zeta) \le a(\xi) - r, \quad b(\zeta) \ge b(\xi). \end{aligned} \tag{3.2}$$

The sets $I_j$ are disjoint in the physical angular space. Suppose also that the comparison sets have overlap at most $N$:

$$\sum_j \mathbf{1}_{J_j}(\xi) \le N \quad \text{for almost every } \xi. \tag{3.3}$$

This is a condition on physical angular sets, not on the number of words. For disjoint angular arms it holds with $N = 1$. Write $C_j = \int_{I_j} c\,d\theta$ and $D_j = \int_{J_j} c\,d\theta$, and assume these numbers are finite. Zero values are allowed.

**Theorem 3.1** (A positive angular-interval comparison)**.** *Let $H(\xi, t)$ be any original Borel selector with $0 \le H \le 1$, and let $\kappa > 0$. Define*

$$\alpha_j = \begin{cases} \min\{1, \kappa D_j/C_j\}, & C_j > 0, \\ 0, & C_j = 0, \end{cases} \qquad T^{\varepsilon}_{H,j}(t) = \int_{I_j} c(\xi)H(\xi,t)\mathbf{1}_{\{a(\xi)<t<b(\xi)\}}\,d\theta.$$

*At every finite collision count and for almost every target roof,*

$$\sum_j \alpha_j T^{\varepsilon}_{H,j}(t) \le \frac{N\kappa}{r}\int_{t-r}^{t+r} \mathsf{T}^{\varepsilon}_{\Delta}(u)\,du. \tag{3.4}$$

*The remaining source is exactly $\sum_j (1-\alpha_j)T^{\varepsilon}_{H,j}$, and satisfies*

$$0 \le \sum_j (1-\alpha_j)T^{\varepsilon}_{H,j}(t) \le \sum_j (C_j - \kappa D_j)_+ \mathbf{1}_{\{T^{\varepsilon}_{1,j}(t)>0\}}. \tag{3.5}$$

*If $C_j \le \kappa D_j$ whenever $C_j > 0$, then the left side of* (3.4) *is the complete selected source on the union of the $I_j$.*

*Proof.* Fix $j$ and a roof $t$ for which $T^{\varepsilon}_{H,j}(t) > 0$. The set of source angles with $a(\xi) < t < b(\xi)$ and positive selected weight then has positive angular measure. By Fubini, (3.2) holds between some such source angle and almost every comparison angle. If $\tau_j = 1$, every $u \in (t, t+r)$ belongs to the corresponding comparison segment; if $\tau_j = -1$, the same assertion holds for $u \in (t-r, t)$. Writing $T_{J_j}(u) = \int_{J_j} c\mathbf{1}_{\{a<u<b\}}\,d\theta$, we obtain

$$rD_j \le \int_{t+\tau_j(0,r)} T_{J_j}(u)\,du, \qquad \alpha_j T^{\varepsilon}_{H,j}(t) \le \alpha_j C_j \le \kappa D_j.$$

When the selected source is zero the desired comparison is immediate. Thus each summand is bounded by the corresponding one-sided integral multiplied by $\kappa/r$. All such intervals lie in $(t-r,t+r)$. Tonelli and (3.3) give $\sum_j T_{J_j}(u) \le N\mathsf{T}^{\varepsilon}_{\Delta}(u)$ and prove (3.4). The remaining source is positive because $0 \le \alpha_j \le 1$. Since $(1-\alpha_j)C_j = (C_j - \kappa D_j)_+$ for $C_j > 0$, its displayed upper bound follows. The case $C_j = 0$ contributes nothing. Countable sums follow from monotone convergence of nonnegative finite sums. No collision limsup has been taken. □

This proof compares integrals of the original positive endpoint minimum. It neither differentiates that minimum nor divides by $a'$, $b'$ or $(1-s)a' + sb'$. There is no flow through a stationary point and no assertion that two prescribed trajectories are joined by an admissible clock. The selector remains at the starting roof; only its positive comparison source is enlarged.

**Corollary 3.2** (The physically summed essential height)**.** *Suppose $N$ and $\kappa$ are uniform in collision count, radius and exact labels. Under* (2.1)*,*

$$\mathcal{H}_M\left(\sum_j \alpha_j T^{\varepsilon}_{H,j}\right) \le CN\kappa\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{3.6}$$

*All widths and constants in this statement are fixed before the collision limit. The estimate permits any number of source intervals and comparison intervals, subject to the stated overlap.*

*Proof.* The right side of (3.4) is $2N\kappa A_r \mathsf{T}^{\varepsilon}_{\Delta}(t)$. Apply the complete paired-average estimate only after the finite-count sum and the pointwise comparison have been formed. □

## 4. Stationary endpoint roofs at a positive critical entrance

The preceding comparison can be verified on a class for which the endpoint speed has no positive lower bound. The condition on the outgoing trace is an integral condition. Neither a derivative of that trace nor a pointwise lower bound for the positive minimum is used.

Consider an oriented angular arm $x = \sigma(\theta - \theta_*) > 0$, $\sigma \in \{-1, 1\}$, on one actual pairing sheet with a regular nongrazing critical entrance. The incoming roof is $a_*$ and, by (2.4), its weighted trace is

$$f^{\varepsilon}_{-}(x) = \frac{N_*}{Q_*(x)}, \qquad \lambda_\circ \le Q_*(x) \le \Lambda_\circ, \qquad N_* \ge 0. \tag{4.1}$$

Suppose the outgoing roof on this arm is

$$b(x) = a_* + d + \psi(x), \qquad 0 \le d \le \delta/2, \tag{4.2}$$

where $\psi$ is continuous, strictly increasing, positive for $x > 0$, and tends to zero at the left endpoint. Assume that its levels $\delta/2, 2\delta, 3\delta$ are reached at points $x_1 < x_2 < x_3$ inside this same physical arm. Set

$$I = (0, x_1), \qquad J = (x_2, x_3).$$

The existence of $x_3$ is an actual angular-reach condition. It is not inferred from an algebraic continuation beyond a true endpoint.

If $N_* > 0$, put

$$w(x) = \min\{1, f^{\varepsilon}_{+}(x)/f^{\varepsilon}_{-}(x)\}, \qquad \mu_J = \frac{1}{|J|}\int_J w(x)\,dx. \tag{4.3}$$

Thus $c^{\varepsilon} = f^{\varepsilon}_{-} w$ exactly. The fraction may vanish on subsets of positive angular measure. If $N_* = 0$, the whole paired source on the arm is zero and is treated separately.

**Lemma 4.1** (Width and positive-trace balance)**.** *For $N_* > 0$ and $\mu_J > 0$,*

$$\frac{\int_I c^\varepsilon dx}{\int_J c^\varepsilon dx} \le \frac{\Lambda_\circ}{\lambda_\circ \mu_J} \frac{x_1}{x_3 - x_2}. \tag{4.4}$$

*If, in addition, $\psi$ is locally absolutely continuous and*

$$\frac{x\psi'(x)}{\psi(x)} \le K \quad \text{for almost every } x \in (x_2, x_3), \qquad K > 0, \tag{4.5}$$

*then*

$$\frac{\int_I c^\varepsilon dx}{\int_J c^\varepsilon dx} \le \frac{\Lambda_\circ K}{\lambda_\circ \mu_J \log(3/2)}. \tag{4.6}$$

*No lower bound on $\psi'$ or bound on $\partial_x \log \psi'$ is part of either assertion.*

*Proof.* On $I$ one has $c^\varepsilon \le N_*/\lambda_\circ$; on $J$, $c^\varepsilon \ge N_* w/\Lambda_\circ$. Integration proves (4.4). Since $\psi \ge 2\delta$ on $[x_2, x_3]$, its logarithm is absolutely continuous there. Hence

$$\log(3/2) = \int_{x_2}^{x_3} \frac{\psi'}{\psi}\, dx \le K \log(x_3/x_2).$$

It follows that

$$x_3 - x_2 \ge x_2\big((3/2)^{1/K} - 1\big) \ge x_1 \log(3/2)/K.$$

Substitution gives (4.6). The numerator $N_*$ cancels; no positive lower bound for its size is required. □

**Theorem 4.2** (A stationary-arm essential-height estimate)**.** *Take any disjoint family of actual angular arms with* (4.1)–(4.2)*. Suppose their comparison intervals satisfy either*

$$\frac{\Lambda_\circ}{\lambda_\circ \mu_J} \frac{x_1}{x_3 - x_2} \le \kappa,$$

*or the sufficient uniform bounds $K \le K_0$ and $\mu_J \ge \mu_0 > 0$ in Lemma 4.1. Let $T^\varepsilon_{H,\mathrm{arm}}$ be any original selected paired source on the union of their source intervals $I$. At finite collision count,*

$$T^\varepsilon_{H,\mathrm{arm}}(t) \le \frac{\kappa}{\delta} \int_t^{t+\delta} \mathsf{T}^\varepsilon_{4\delta}(u)\, du \quad \text{for almost every } t. \tag{4.7}$$

*Consequently*

$$\mathcal{H}_M(T^\varepsilon_{H,\mathrm{arm}}) \le C\kappa\{\varepsilon^{1/16} + (1 + \varepsilon^{-1})\delta\}, \qquad \kappa = \frac{\Lambda_\circ K_0}{\lambda_\circ \mu_0 \log(3/2)} \tag{4.8}$$

*in the sufficient case. Every original relative position $0 < s < 1$ is included. The family may have arbitrarily many physical words.*

*Proof.* For a source angle $x \in I$ the residence is at most $d + \delta/2 \le \delta$. For a comparison angle $y \in J$, it is less than $d + 3\delta \le 7\delta/2 < 4\delta$. Their incoming roofs are equal, and

$$b(y) - b(x) > 2\delta - \delta/2 = 3\delta/2 > \delta.$$

Thus the positive inclusion (3.2) holds with $r = \delta$. Lemma 4.1 proves the mass balance. Different arms are disjoint in the original physical angular space; therefore their comparison intervals have overlap one, even if geometrical boundary points coincide on different incident sheets. The proof of Theorem 3.1, with every sign positive, gives (4.7). Its right side is at most $2\kappa A_\delta \mathsf{T}^\varepsilon_{4\delta}$. The inherited paired-average estimate proves (4.8). Arms with $N_* = 0$ have no positive minimum and add zero. □

The arm assumption, including the weighted fraction in (4.3), is imposed on actual endpoint traces. It is weaker than a positive pointwise lower bound or logarithmic variation bound on their minimum. Failure of this average condition is quantified by the positive deficit in (3.5), not assigned zero mass.

4.1. **Paying a deficient comparison trace by the original volume.** The average fraction in (4.3) need not be bounded below when the original weighted trace variation is available. Its possible loss has a direct pointwise payment. Retain the actual uncut source $v^0$ from (2.2).

**Theorem 4.3** (Stationary-arm height without a trace-fraction hypothesis)**.** *On a disjoint family of the actual critical arms above, assume only* (4.1)–(4.2) *and a uniform geometric width ratio*

$$\frac{\Lambda_\circ}{\lambda_\circ}\frac{x_1}{x_3-x_2}\le\kappa.$$

*There is no lower bound on $\mu_J$. For every original selector $0\le H\le 1$, at finite count and almost every target roof,*

$$T^{\varepsilon}_{H,\mathrm{arm}}(t)\le\frac{\kappa}{\delta}\int_t^{t+\delta}\mathsf{T}^{\varepsilon}_{4\delta}(u)\,du + C\kappa(1+\varepsilon^{-1})\int_{t-\delta}^{t+4\delta}v^0(u)\,du. \tag{4.9}$$

*The positive comparison deficit itself obeys*

$$\sum_j(1-\alpha_j)T^{\varepsilon}_{H,j}(t)\le C\kappa(1+\varepsilon^{-1})\int_{t-\delta}^{t+4\delta}v^0(u)\,du. \tag{4.10}$$

*Consequently the entire selected arm source satisfies*

$$\mathcal{H}_M(T^{\varepsilon}_{H,\mathrm{arm}})\le C\kappa\{\varepsilon^{1/16}+(1+\varepsilon^{-1})\delta\}. \tag{4.11}$$

*The sufficient profile condition* (4.5) *gives $\kappa=\Lambda_\circ K_0/(\lambda_\circ\log(3/2))$ for $K\le K_0$. No source endpoint-speed, incidence, residence or relative-position lower cutoff is imposed.*

*Proof.* For one arm set

$$P_j=\int_{J_j}f^{\varepsilon}_-(x)\,dx,\qquad E_j=\int_{J_j}[f^{\varepsilon}_-(x)-f^{\varepsilon}_+(x)]_+\,dx.$$

The positive minimum gives the exact identity $D_j=P_j-E_j$. The critical incoming formula and the geometric width ratio imply $C_j\le\kappa P_j$. Therefore

$$(C_j-\kappa D_j)_+\le\kappa E_j. \tag{4.12}$$

This is valid even when $D_j=0$. Arms with $N_*=0$ again contribute zero; no division by a vanishing incoming trace is used.

The retained derivative bound for the original weighted density, integrated on the same comparison segment, yields

$$[f^{\varepsilon}_-(x)-f^{\varepsilon}_+(x)]_+\le C(1+\varepsilon^{-1})\int_{a_j}^{b(x)}A(u,x)I_m(u,x)/z(u,x)\,du.$$

One may first integrate on compact interior subintervals and pass to the one-sided traces. The uncut volume input and Tonelli justify the positive limiting integral. This pays a possible decrease of the trace; it does not assume that the two traces are comparable.

At a roof with $T^{\varepsilon}_{H,j}(t)>0$, some original source segment on $I_j$ contains $t$. Since every such segment has length at most $\delta$ and has the same incoming roof $a_j$, one has $t-\delta<a_j<t$. Each comparison segment on $J_j$ has length less than $4\delta$. Its entire interior consequently lies in

$(t-\delta, t+4\delta)$. The comparison angular sets are disjoint on the original physical sheet space. After their interiors are summed, the Borel segment identity bounds their combined variation by the integral of the same complete $v^0$ over this one roof window, with multiplicity one. Equation (4.12) and (3.5) prove (4.10). Adding the paid comparison in Theorem 3.1 gives (4.9). The first term uses only positive roof translates, since the incoming roof is constant on each arm.

Finally take the collision limsup with $\delta, \varepsilon, \kappa$ fixed. The paired-average input estimates the first integral. The uncut volume input estimates the second by $C\kappa(1+\varepsilon^{-1})\delta$. Auxiliary spectral bands are enlarged only afterwards. This proves (4.11). □

Thus poor outgoing trace balance on these arms is not assigned to an uncontrolled positive remainder: its entire shortage is estimated at the original target roof by (4.10). What still requires physical information is the level reach and width geometry, and coverage of the complete remaining paired family by such arms or other comparisons.

**Corollary 4.4** (Finite-order stationary profiles)**.** *Suppose on an actual arm*

$$\psi(x) = \beta x^k(1+O(x^\eta)), \qquad \psi'(x) = k\beta x^{k-1}(1+O(x^\eta)), \qquad \beta, \eta > 0, \quad k \geq 1.$$

*For all sufficiently small widths on that arm,* (4.5) *holds with* $K = k+1$*. Provided the required physical reach holds, Theorem 4.3 applies also for* $k > 1$*, when the endpoint speed vanishes at the arm's initial point. No uniform bound on physical orders* $k$*, on the size of the asymptotic neighborhood or on comparison reach is inferred from finite-count definability.*

*Proof.* The displayed asymptotics give $x\psi'/\psi \to k$. The assertion then follows from Lemma 4.1 and Theorem 4.3. □

4.2. **Recovery can occur when every preceding clock is small.** This elementary pairing-sheet calculation distinguishes the new comparison from admission by any of the three previous flows. It is not asserted to be another physical Lorentz realization. Set $a = 0$, $b(x) = x^k$, $c = 1$, $k \geq 2$, and restrict the comparison collar to $0 < \ell < 4\delta$. The incoming-distance clock is zero. For the outgoing-distance flow at positive roof time $h$,

$$x_h = (x^k + h)^{1/k}, \qquad J_h^+ = \left(\frac{x}{x_h}\right)^{k-1}.$$

Its forward contribution for $0 < h < \delta$ is at most

$$\frac{kx^{k-1}}{\delta}\big((x^k+\delta)^{1/k} - x\big). \tag{4.13}$$

The negative-time contribution is at most $x^k/\delta$, because the starting gap must remain smaller than the new residence. At fixed relative position $s$, the relative clock has total roof reach at most $4s\delta$, so its capped value is at most $4s$.

For any fixed $q > 0$, take $s < q/8$ and then take $x/\delta^{1/k}$ sufficiently small. Both terms in (4.13) and its negative-time bound tend to zero; all three old clocks are below $q$ on a set of positive paired measure. Nevertheless the intervals

$$I = (0, (\delta/2)^{1/k}), \qquad J = ((2\delta)^{1/k}, (3\delta)^{1/k})$$

satisfy the positive comparison, with $C_I/D_J \leq k/\log(3/2)$. The new inequality controls their entire source, including that positive-measure clock-failure set. Its proof uses neither a velocity bound on $I$ nor passage of a flow through $x = 0$.

## 5. The recovered source and the original pointwise theorem

Retain the original schedule in (2.5). Let $H_B$ be exactly the original selector for the source which fails all three preceding clocks, so that $\mathsf{U}_B^{\mathrm{rem}} = T_{H_B}^{\varepsilon(B)}$. This selector includes the original short-source test and is supported on $\ell \le \delta(B)$. It is not redefined by the present comparison.

Choose comparison pairs as in Theorem 3.1, with $r = \delta$, $\Delta = 4\delta$ and constants $N_B, \kappa_B$ fixed before the collision limit. They may, for example, be the disjoint critical arms of Theorem 4.2. The exact split is

$$\begin{aligned} \mathsf{V}_B^{\mathrm{int}}(t) &= \sum_j \alpha_j T_{H_B,j}^{\varepsilon(B)}(t), \\ \mathsf{R}_B^{\mathrm{int}}(t) &= T_{H_B \mathbf{1}_{(\cup_j I_j)^c}}^{\varepsilon(B)}(t) + \sum_j (1-\alpha_j) T_{H_B,j}^{\varepsilon(B)}(t). \end{aligned} \tag{5.1}$$

Hence $\mathsf{U}_B^{\mathrm{rem}} = \mathsf{V}_B^{\mathrm{int}} + \mathsf{R}_B^{\mathrm{int}}$ at finite count, with both terms nonnegative. The second sum is an explicitly weighted shortage of comparison trace, with upper bound (3.5). The uncovered source in the first term is retained in full.

**Theorem 5.1** (Positive recovery at the original outer schedule)**.** *Suppose $N_B\kappa_B \le C\varepsilon(B)^{-1/32}$ uniformly in collision count, radius and exact labels. Then*

$$\mathcal{H}_M(\mathsf{V}_B^{\mathrm{int}}) \le C_M B^{-1/384}, \qquad \mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{int}}) + C_M B^{-1/384}. \tag{5.2}$$

*If $N_B\kappa_B$ is bounded independently of $B$, the first bound improves to $C_M B^{-1/192}$. Along the same schedule the original pointwise assertion is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{int}}) \to 0$.*

*Proof.* Apply Corollary 3.2 to the exact starting selector $H_B$. Since $\delta = O(\varepsilon^4)$,

$$\mathcal{H}_M(\mathsf{V}_B^{\mathrm{int}}) \le CN_B\kappa_B\{\varepsilon^{1/16} + (1+\varepsilon^{-1})4\delta\} \le C(\varepsilon^{1/32} + \varepsilon^{95/32}).$$

This is $O(B^{-1/384})$; for bounded $N_B\kappa_B$ the leading power is $\varepsilon^{1/16} = O(B^{-1/192})$. Insert the exact positive split (5.1) into the inherited scalar bound (2.6). This proves sufficiency. Conversely, the retained necessity statement gives $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{rem}}) \to 0$ under scalar closure. Since $0 \le \mathsf{R}_B^{\mathrm{int}} \le \mathsf{U}_B^{\mathrm{rem}}$, necessity follows. The complete physical sums precede the height functional in every step. □

For the critical-arm theorem a uniform $K_0$ and $\mu_0 > 0$ give the stronger rate for the entire selected family. More generally, any outer choice with $K_B/\mu_B \le C\varepsilon(B)^{-1/32}$ gives the first rate. This allows an expanding family of finite-order profiles, but is not a claim that their order or trace fraction is uniformly controlled on all physical pairs. Every such threshold is chosen from $B$, never from the collision count.

**Corollary 5.2** (Removing the whole stationary-critical arm contribution)**.** *Suppose the comparison pairs are the disjoint actual critical arms of Theorem 4.3, with $\kappa_B \le C\varepsilon(B)^{-1/32}$. Then*

$$\mathcal{H}_M\left(\sum_j (1-\alpha_j) T_{H_B,j}^{\varepsilon(B)}\right) \le C\varepsilon(B)^{95/32}.$$

*The complete original source on these arms, including its entire three-clock-failure part, has height $O(B^{-1/384})$. With a bounded geometric ratio the latter rate is $O(B^{-1/192})$. In particular, setting $\mathsf{U}_B^{\mathrm{outside}} = T_{H_B \mathbf{1}_{(\cup_j I_j)^c}}^{\varepsilon(B)}$, one has*

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B^{\mathrm{outside}}) + C_M B^{-1/384}. \tag{5.3}$$

*Along this schedule the original pointwise assertion is equivalent to $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{outside}}) \to 0$.*

*Proof.* Apply (4.10) on the original starting selector. The complete uncut volume estimate gives $C\kappa_B(1+\varepsilon^{-1})\delta = O(\varepsilon^{95/32})$. Adding this paid shortage to (5.2) removes the entire angular-arm source, not only a fraction admitted by trace balance. The exact decomposition into that source and its positive outside complement proves (5.3). Necessity follows from domination by the original $\mathsf{U}_B^{\mathrm{rem}}$, and sufficiency from the new estimate. No count-dependent width is chosen. □

5.1. **The remaining physical estimate.** The new comparison has an actual essential-height conclusion on the positive recovered source. It does not turn an average, a signed boundary current or a finite local norm into the target theorem. Stationary endpoint speed and rapid pointwise variation of the minimum are no longer obstructions on the interval families just estimated. The model calculation shows that this recovered family need not have been admitted by any earlier clock.

Two separate matters are still not proved on the complete Lorentz family: coverage by comparison intervals with a schedule-compatible physical overlap and mass balance, and vanishing of the uncovered source. On the critical arms of Theorem 4.3, the entire comparison-trace shortage is already paid by the original uncut volume. In particular a flat endpoint pulse on an entire isolated chamber need not have any comparison interval satisfying (3.2); its positive mass is not removed. Loss of physical continuation before the comparison levels, and singular critical extensions remain in the outside source. General comparison families without the critical incoming normalization may still have a trace shortage in $\mathsf{R}_B^{\mathrm{int}}$.

Thus (5.2) is not a proof that the full remaining source has vanishing height. The exact-label pointwise theorem remains the governing assertion (1.5); its full closure and the corresponding unrestricted point-conditioned consequences are not promoted. The original arithmetic zero classes remain in the unnormalized theorem, and scalar closure still precedes any denominator, same-roof ratio or unrestricted likelihood statement.

## 6. Complete positive-residence persistence without comparison corridors

Throughout, $(a(\xi), b(\xi))$ is the actual maximal regular roof segment on a disjoint physical pairing sheet, $\ell(\xi) = b(\xi) - a(\xi)$, and $c^{\varepsilon}(\xi)$ is its unchanged positive endpoint-trace minimum. All exact labels, the original section and the original event are retained. For an arbitrary Borel selector $0 \le H(\xi, t) \le 1$ set

$$T^{\varepsilon}_{H,[h,\Delta)}(t) = \int c^{\varepsilon}(\xi) H(\xi, t) \mathbf{1}_{\{h \le \ell(\xi) < \Delta\}} \mathbf{1}_{\{a(\xi) < t < b(\xi)\}}\, d\theta(\xi).$$

**Theorem 6.1** (Original source persistence at every positive residence)**.** *At each finite collision count, for every $0 < h < \Delta$ and almost every original roof $t$,*

$$0 \le T^{\varepsilon}_{H,[h,\Delta)}(t) \le \frac{1}{h} \int_{t-h}^{t+h} \mathsf{T}^{\varepsilon}_{\Delta}(u)\, du = 2A_h \mathsf{T}^{\varepsilon}_{\Delta}(t). \tag{6.1}$$

*Consequently, for every previously fixed positive $h, \Delta, \varepsilon$,*

$$\mathcal{H}_M(T^{\varepsilon}_{H,[h,\Delta)}) \le C\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{6.2}$$

*The leading constant is independent of the previously fixed $h$. The conclusion covers all physical endpoint types within the inherited paired source: regular and nonstationary pairs, flat endpoint pulses, reach-deficient arms and singular extensions. No positive interval comparison or endpoint derivative is required.*

*Proof.* If $a < t < b$ and $\ell = b - a \ge h$, put $d_- = t - a$ and $d_+ = b - t$. Then

$$\int_{t-h}^{t+h} \mathbf{1}_{\{a<u<b\}}\, du = \min(h, d_-) + \min(h, d_+) \ge h$$

because $d_- + d_+ \geq h$. Multiply this scalar inequality by $c^\varepsilon(\xi)H(\xi,t)$ and sum over the actual Borel physical sheets. The selector remains at the original target roof and is neither differentiated nor transported. Tonelli permits enlargement of the positive right-hand source to the complete $\mathsf{T}^\varepsilon_\Delta$. This proves the finite-count inequality before taking any collision limit. Equation (2.1), used only with the fixed window $h$, gives the ordered essential-height estimate. No multiplicity or angular word count enters the argument. □

**Corollary 6.2** (Physical flat pulses at positive residence)**.** *For every $B$ choose $h_B \in (0, \delta(B))$ from the original outer parameters, before the collision limit. The selected original source on all regular pairing segments $h_B \leq \ell \leq \delta(B)$, including every positive-residence flat-pulse sheet even when no comparison interval exists, has ordered height $O(B^{-1/192})$, uniformly in the choice of previously fixed $h_B$.*

*Proof.* Use $\Delta = 4\delta(B)$ in Theorem 6.1, and the original schedule $\delta(B) = O(\varepsilon(B)^4)$, $\varepsilon(B) = A_0 B^{-1/12}$. The first exponent is $B^{-1/192}$ and the second contribution decays faster. □

The quantifier is important. A leading constant uniform over positive *fixed* windows does not imply a finite-count estimate at $h = 0$, nor may a window shrink with collision count. The complete microscopic residence source remains positive and is isolated below.

## 7. Adaptive positive comparison at the actual reached scale

The preceding stationary-arm argument requires physical reach to the outgoing levels $\delta/2, 2\delta, 3\delta$. On the same original regular backward-critical arm, retain the incoming roof $a_*$ and positive incoming trace

$$f^\varepsilon_-(x) = N_*/Q_*(x), \qquad \lambda_\circ \leq Q_*(x) \leq \Lambda_\circ,$$

and let $b(x) = a_* + d + \psi(x)$ with $0 \leq d \leq \delta/2$. Choose an *outer, collision-independent* scale $0 < \rho \leq \delta$. Assume $\psi$ is continuous and strictly increasing and *actually reaches*, before a true physical boundary, the levels

$$\psi(x_1) = \rho/8, \qquad \psi(x_2) = \rho/2, \qquad \psi(x_3) = 3\rho/4, \quad x_1 < x_2 < x_3. \tag{7.1}$$

Set $I = (0, x_1)$ and $J = (x_2, x_3)$ on this very physical sheet.

**Theorem 7.1** (Smaller-reach critical-arm source with paid deficit)**.** *Suppose the comparison intervals of a family of such actual arms are disjoint on the original incident-component pairing-sheet space. Assume their geometric ratios satisfy*

$$\frac{\Lambda_\circ}{\lambda_\circ}\frac{|I|}{|J|} \leq \kappa$$

*with a common outer cost $\kappa > 0$, independent of the collision count and exact labels. For every original Borel selector $H$, at finite collision count and almost every roof $t$,*

$$T^\varepsilon_{H,\mathrm{reach}}(t) \leq \frac{\kappa}{\rho/4}\int_t^{t+\rho/4} \mathsf{T}^\varepsilon_{4\delta}(u)\,du + C\kappa(1+\varepsilon^{-1})\int_{t-\delta}^{t+4\delta} v^0(u)\,du. \tag{7.2}$$

*Thus*

$$\mathcal{H}_M(T^\varepsilon_{H,\mathrm{reach}}) \leq C\kappa\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\delta\}. \tag{7.3}$$

*This is the complete source on the selected reached arms, not only a trace-balanced fraction; no endpoint-speed lower bound or outgoing trace-fraction bound is imposed.*

*Proof.* For $x \in I$, $\ell(x) = d + \psi(x) \le 5\delta/8 < \delta$; for $y \in J$, $\ell(y) < d + 3\rho/4 \le 5\delta/4 < 4\delta$. Moreover $a(y) = a(x) = a_*$ and $b(y) - b(x) > 3\rho/8 > \rho/4$. Therefore the actual one-sided inclusion (3.2) holds at $r = \rho/4$. For each arm put

$$C_j = \int_I c^\varepsilon, \quad P_j = \int_J f_-^\varepsilon, \quad D_j = \int_J c^\varepsilon, \quad E_j = \int_J [f_-^\varepsilon - f_+^\varepsilon]_+.$$

The incoming critical trace and the geometric ratio give $C_j \le \kappa P_j$, while the positive minimum gives $D_j = P_j - E_j$. Hence $(C_j - \kappa D_j)_+ \le \kappa E_j$, even when $D_j = 0$. The positive comparison theorem 3.1 pays the admitted fraction on $(t, t + \rho/4)$. For any original source segment containing $t$, one has $a_* < t < a_* + \delta$, while every comparison segment lies in $(a_*, a_* + 4\delta)$. The inherited weighted one-sided trace-variation estimate (2.3), integrated over the actual comparison segments, pays the full $E_j$ in $(t - \delta, t + 4\delta)$. Disjointness of the physical arms gives multiplicity one in the uncut $v^0$ source sum. This proves (7.2) before any limit. With $\rho, \delta, \kappa, \varepsilon$ fixed before collision limsup, apply (2.1) and (2.2) to the two fixed windows. □

**Corollary 7.2** (Elastic width criterion)**.** *If the actual $\psi$ has $x\psi'(x)/\psi(x) \le K_B$ almost everywhere on $(x_2, x_3)$, then $\kappa \le \Lambda_\circ K_B/(\lambda_\circ \log(3/2))$ is admissible. For $K_B \le C\varepsilon(B)^{-1/32}$, the complete reached source has height $O(B^{-1/384})$. The conclusion permits $\rho_B \ll \delta(B)$ and physical arms which never reach the old level $3\delta(B)$.*

*Proof.* Integrate $\psi'/\psi$ across the two actually reached levels: $\log(3/2) \le K_B \log(x_3/x_2)$, so $x_3 - x_2 \ge x_1 \log(3/2)/K_B$. Substitute in (7.3). □

No theorem here asserts that an arm reaching less than $3\rho/4$ can be continued past its true physical boundary. Its original positive source remains in the complementary class.

## 8. Exact microscopic remainder of the original positive source

Retain the complete original short-source selector $H_B$ of (2.5). Let $\mathscr{C}_B$ be a Borel union of disjoint physical critical-arm source sets already paid by Corollary 5.2, and let $\mathscr{D}_B$ be disjoint source sets satisfying Theorem 7.1, after first removing any membership in $\mathscr{C}_B$. Both families have the corresponding actual reach, physical overlap and outer balance costs. Choose $0 < h_B < \delta(B)$ using only the outer parameter $B$. Define the nonnegative *original* microscopic restriction

$$\mathsf{U}_B^{\mathrm{micro}}(t) = T^{\varepsilon(B)}_{H_B \mathbf{1}_{(\mathscr{C}_B \cup \mathscr{D}_B)^c} \mathbf{1}_{\{0 < \ell < h_B\}}}(t). \tag{8.1}$$

This retains all endpoint types, original section tests, occupation and displacement labels, the actual roof, the guard and the physical normalization.

**Theorem 8.1** (Exact positive scalar reduction at any fixed resolution)**.** *Assume that the old and adaptive selected families meet their actual physical hypotheses and that their uniform outer costs are at most $C\varepsilon(B)^{-1/32}$. Then, at the unchanged outer schedule, for every such $h_B$ fixed before collision limsup,*

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) + C_M B^{-1/384}. \tag{8.2}$$

*The unrestricted original exact-label raw pointwise theorem is equivalent to*

$$\mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) \longrightarrow 0 \qquad (B \longrightarrow \infty).$$

*The microscopic term includes all flat pulses, true reach failures, singular critical extensions, nonstationary sheets and unbalanced comparison deficits which have not otherwise been paid; none is deleted or relabeled as a signed boundary current.*

*Proof.* At each finite collision count, disjointization by first physical membership yields the exact nonnegative identity

$$\begin{aligned}\mathsf{U}_B^{\mathrm{rem}} = {} & T^{\varepsilon(B)}_{H_B\mathbf{1}_{\mathscr{C}_B}} + T^{\varepsilon(B)}_{H_B\mathbf{1}_{\mathscr{D}_B}} \\ & + T^{\varepsilon(B)}_{H_B\mathbf{1}_{(\mathscr{C}_B\cup\mathscr{D}_B)^c}\mathbf{1}_{\{\ell\ge h_B\}}} + \mathsf{U}_B^{\mathrm{micro}}.\end{aligned}$$

The first term is paid by the inherited complete critical-arm comparison; the second by Theorem 7.1. The third is paid by the full-source persistence estimate 6.1, irrespective of its physical endpoint class, because $H_B$ already selects $\ell \le \delta(B)$. The first two ordered heights are $O(B^{-1/384})$, and the third is $O(B^{-1/192})$. Apply subadditivity and the inherited scalar bound (2.6) to obtain (8.2). For necessity use the inherited scalar equivalence and $0 \le \mathsf{U}_B^{\mathrm{micro}} \le \mathsf{U}_B^{\mathrm{rem}}$. All widths and costs are fixed before collision limsup. No cutoff is removed at a later collision count. □

**Proposition 8.2** (A sufficient untruncated residence insertion)**.** *Suppose for some $\sigma > 0$ and every fixed $M$, the following* additional, not presently verified *original-source bound is available with a constant independent of $B$, collision count, radius and exact labels:*

$$\sup_{B\ge B_0}\limsup_{m}\sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \int c^{\varepsilon(B)}(\xi)H_B(\xi,t)\ell(\xi)^{-\sigma}\,\mathbf{1}_{\{0<\ell\le\delta(B)\}}\mathbf{1}_{\{a<t<b\}}\,d\theta(\xi) < \infty. \tag{8.3}$$

*Then the unchanged exact-label raw pointwise theorem follows. The proposition is an implication; no estimate of the form* (8.3) *is claimed to have been proved for the full Lorentz source.*

*Proof.* On $0 < \ell < h_B$, $1 \le h_B^\sigma \ell^{-\sigma}$. Consequently $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) \le C_M h_B^\sigma$. Choose $h_B \downarrow 0$ from the outer schedule; then apply Theorem 8.1. This is the correct finite-count source insertion, not a post-limit interpolation of fixed-width bounds. □

The task still to be proved is the vanishing of the original positive shrinking-residence height, or a genuine uniform finite-count source insertion strong enough to imply it. No unrestricted denominator, same-roof conditional path or forward likelihood conclusion is promoted before that scalar proof.

## 9. A logarithmic microscopic-residence criterion

The following criterion concerns the *original positive paired measure*, not an auxiliary volume collar or an integrated surrogate. It isolates a strictly weaker sufficient hypothesis than any positive inverse-power residence insertion while retaining the order of the collision and microscopic limits.

For a fixed outer band $B$ and fixed original exact labels, let $d\pi^{\varepsilon,+}_{m,R,n,k}$ denote the nonnegative paired physical source of the preceding theorem and let $\ell > 0$ be its actual maximal regular residence. We use exactly the same original pushforward to the roof as in the definition of $\mathsf{U}_B^{\mathrm{micro}}$. All integrals below are with respect to this same source, including its physical normalization. Fix a deterministic scale $h_0 > 0$, independent of the collision count, and define

$$L_{h_0}(\ell) = \log(eh_0/\ell) \quad (0 < \ell \le h_0).$$

Write $\mathsf{W}_{\gamma,h_0}$ for the roof pushforward of $L_{h_0}(\ell)^\gamma \mathbf{1}_{\{0<\ell\le h_0\}}\,d\pi^{\varepsilon,+}$ restricted to the *same unmatched original physical source* as $\mathsf{U}_B^{\mathrm{micro}}$.

**Theorem 9.1** (Positive microscopic localization by logarithmic insertion)**.** *For every $\gamma > 0$ and every $0 < h < h_0$ one has, at every finite collision count and for every original exact label, the pointwise inequality of positive roof densities*

$$0 \le \mathsf{U}_B^{\mathrm{micro}}(t;h) \le \big[\log(eh_0/h)\big]^{-\gamma}\mathsf{W}_{\gamma,h_0}(t).$$

*Here* $\mathsf{U}_B^{\mathrm{micro}}(t;h)$ *is the actual unmatched positive source on* $0<\ell<h$*; the restriction is made before the roof pushforward and no original test is dropped. Consequently, if at the fixed* $B,h_0,\gamma$ *one establishes the* collision-uniform original-source insertion estimate

$$K_{B,M,\gamma,h_0}:=\limsup_{m\to\infty}\sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \mathsf{W}_{\gamma,h_0}(t)<\infty,$$

*then*

$$\mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}(\cdot;h))\le K_{B,M,\gamma,h_0}[\log(eh_0/h)]^{-\gamma}.$$

*In particular the ordered microscopic source height vanishes as* $h\downarrow 0$ *at fixed* $B$*,* provided the displayed insertion estimate is proved for the original Lorentz source*. No such insertion estimate is asserted here.*

*Proof.* On $0<\ell<h$, monotonicity of the logarithm gives $L_{h_0}(\ell)^\gamma\ge\log(eh_0/h)^\gamma$. Multiply by the original nonnegative paired density and retain every endpoint, section, occupation and displacement test. Pushing forward the resulting inequality by the same measurable roof map preserves order without any Jacobian substitution or count of collision words. This establishes the first claim at finite count. Essential supremum, original-label supremum and collision limsup are monotone and positively homogeneous, proving the second claim. The logarithmic factor vanishes only after this ordered bound, so no collision-dependent cutoff or spectral band has been introduced. □

*Remark* 9.2 (Strict weakness and its limitation). For every $\sigma>0$ and $\gamma>0$, the function $x^\sigma[\log(e/x)]^\gamma$ is bounded on $(0,1]$. Therefore, on $(0,h_0]$, an estimate for the genuine inverse-power insertion $(h_0/\ell)^\sigma$ implies the logarithmic hypothesis, up to a constant depending on $\sigma,\gamma$. The reverse implication does not hold for arbitrary positive measures: a mass concentrated at $\ell_j=h_0e^{-j}$ with weight proportional to $j^{-\gamma-2}$ has a finite logarithmic $\gamma$-moment and an infinite inverse-power $\sigma$-moment for every $\sigma>0$. Thus the logarithmic criterion is a mathematically weaker route to the same unaltered pointwise endpoint. Neither criterion is proved for the complete original Lorentz source here.

## 10. Two-ended comparison on moving physical roofs

Throughout this section the collision count, radius, exact labels and $\varepsilon>0$ are fixed. All angular integrals are taken on the original disjoint space $\Theta_{\mathrm{phys}}$ of incident pairing sheets. Put

$$\begin{aligned} f_-(\theta)&=f^\varepsilon(a+,\theta), & f_+(\theta)&=f^\varepsilon(b-,\theta),\\ p_\vee(\theta)&=\max\{f_-(\theta),f_+(\theta)\}, & e_\partial(\theta)&=|f_-(\theta)-f_+(\theta)|. \end{aligned} \tag{10.1}$$

Thus $p_\vee=c^\varepsilon+e_\partial$. The traces used below are the finite one-sided traces of the inherited physical source. No critical-point formula is imposed.

**Lemma 10.1** (Endpoint passage)**.** *On a regular segment where* $f^\varepsilon$ *is locally absolutely continuous and has the stated one-sided traces,*

$$e_\partial(\theta)\le C(1+\varepsilon^{-1})\int_{a(\theta)}^{b(\theta)} A(u,\theta)I_m(u,\theta)/z(u,\theta)\,du. \tag{10.2}$$

*Proof.* On each compact subinterval the physical density $f^\varepsilon$ is absolutely continuous. Integrate (2.3) between two interior points, then let the points tend monotonically to the two ends. The left side converges by the one-sided trace convention and the right side by monotone convergence. If that integral is infinite, the asserted nonnegative inequality is still valid. Neither the minimum, the maximum, nor a target selector has been differentiated. □

**Definition 10.2** (Physical comparison family)**.** Fix $0 < r < \Delta$, $0 < N < \infty$ and $0 < \kappa < \infty$. Let $I_j$ be countably many disjoint Borel angular source sets, let $0 \le s_j \le 1$ be supported on $I_j$, and let $0 \le w_j \le 1$ be Borel comparison weights. For each $j$ choose $\tau_j \in \{-1, 1\}$. The following conditions refer to actual maximal physical segments:

$$0 < b(\eta) - a(\eta) < \Delta \qquad \text{on } \{w_j > 0\}, \tag{10.3}$$

$$\sum_j w_j(\eta) \le N \qquad \text{for almost every } \eta \in \Theta_{\mathrm{phys}}. \tag{10.4}$$

For product-almost every $(\xi, \eta) \in I_j \times \{w_j > 0\}$ require

$$(a(\xi), b(\xi)) + \tau_j[0, r] \subset (a(\eta), b(\eta)). \tag{10.5}$$

Here product-almost everywhere uses angular measure in the first variable and $w_j(\eta)\, d\eta$ in the second. Define the finite nonnegative capacities

$$\begin{aligned} C_j &= \int_{I_j} s_j c^{\varepsilon}\, d\theta, \quad P_j = \int w_j p_{\vee}\, d\theta, \\ D_j &= \int w_j c^{\varepsilon}\, d\theta, \quad E_j = \int w_j e_{\partial}\, d\theta. \end{aligned} \tag{10.6}$$

In particular $P_j = D_j + E_j$. The same fixed exact labels are used on the source and comparison sides.

For a Borel selector $0 \le H \le 1$, held at its original target roof, write

$$T_j(t) = \int_{I_j} s_j(\theta) c^{\varepsilon}(\theta) \mathbf{1}_{\{a(\theta) < t < b(\theta)\}} H(\Psi(\theta, t))\, d\theta.$$

Weights $s_j, w_j$ do not depend on the target roof. In contrast, $H$ may be any of the original roof-dependent failure selectors.

**Lemma 10.3** (Common window without a common entrance)**.** *If $T_j(t) > 0$, then for $w_j\, d\eta$-almost every comparison sheet*

$$t + \tau_j[0, r] \subset (a(\eta), b(\eta)) \subset (t - \Delta, t + \Delta). \tag{10.7}$$

*Proof.* By Fubini, outside a fixed angular null set of source points, (10.5) holds for almost every comparison point. Positivity of $T_j(t)$ implies that the source points whose segment contains $t$ and whose integrand is positive form a set of positive angular measure. It therefore contains a point outside that null set. Apply the interval inclusion at this point. Since the comparison segment contains $t$ and has length less than $\Delta$, both its ends lie within distance $\Delta$ of $t$. For a countable family the angular exceptional sets can be combined before choosing $t$. The inherited coarea identities are then used on their common roof versions at fixed $m, R, n, k$. This does not assert a null set uniform over all radii. □

**Theorem 10.4** (Two-ended positive-envelope estimate)**.** *Let Definition 10.2 hold and suppose*

$$C_j \le \kappa P_j \qquad \text{for every } j. \tag{10.8}$$

*Then, on the original roof kernels at finite count,*

$$\begin{aligned} \sum_j T_j(t) \le \frac{N\kappa}{r} \int_{t-r}^{t+r} \mathsf{T}^{\varepsilon}_{\Delta}(u)\, du \\ + CN\kappa(1 + \varepsilon^{-1}) \int_{t-\Delta}^{t+\Delta} v^0(u)\, du. \end{aligned} \tag{10.9}$$

*If the hypotheses hold uniformly with previously fixed parameters,*

$$\mathcal{H}_M\Big(\sum_j T_j\Big) \le CN\kappa\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{10.10}$$

*There is no lower bound on $D_j/P_j$ and neither endpoint roof is required to be stationary.*

*Proof.* Blocks with $C_j = 0$ contribute nothing. On every other block set $\alpha_j = \min\{1, \kappa D_j/C_j\}$. The balance and $P_j = D_j + E_j$ give the exact algebraic bound

$$(1-\alpha_j)C_j = (C_j - \kappa D_j)_+ \le \kappa E_j. \tag{10.11}$$

At an active roof $t$, Lemma 10.3 gives

$$rD_j \le \int_{t+\tau_j(0,r)} \int w_j(\eta) c^\varepsilon(\eta) \mathbf{1}_{\{a(\eta)<u<b(\eta)\}}\, d\eta\, du.$$

Together with $T_j(t) \le C_j$ this bounds $\alpha_j T_j(t)$. Sum over active blocks, extend both oriented windows to $(t-r, t+r)$, and use (10.3)–(10.4). The result is the first term in (10.9).

For the complementary part use (10.11), then Lemma 10.1. By Lemma 10.3 the entire variation integral for every active comparison sheet lies in $(t-\Delta, t+\Delta)$. Consequently Tonelli's theorem and physical overlap give

$$\sum_{j:T_j(t)>0} E_j \le CN(1+\varepsilon^{-1}) \int_{\Theta_{\mathrm{phys}}} \int_{t-\Delta}^{t+\Delta} \mathbf{1}_{\{a(\eta)<u<b(\eta)\}} A(u,\eta) I_m(u,\eta)/z(u,\eta)\, du\, d\eta.$$

The last physical integral is the original uncut volume integral in (2.2). This proves the second term. For a countable family first sum over finitely many blocks and increase that family; all terms are nonnegative and the same global bound applies. No limit superior is interchanged with that sum.

Finally the first term of (10.9) is $2N\kappa A_r \mathsf{T}^\varepsilon_\Delta(t)$ and the second is $2CN\kappa(1+\varepsilon^{-1})\Delta A_\Delta v^0(t)$. Apply (2.1) and (2.2) at those previously fixed widths. □

**Corollary 10.5** (Nested moving roofs)**.** *On an actual angular arm suppose $a(x)$ is nonincreasing and $b(x)$ is nondecreasing. Let $I = (0, x_1)$ and $J = (x_2, x_3)$, where $x_1 < x_2 < x_3$ are attained before any true physical boundary. Suppose $b(x_2) \ge b(x_1) + r$, comparison lengths are below $\Delta$, and $\int_I c^\varepsilon \le \kappa \int_J p_\vee$. Then this arm satisfies the positive-orientation hypotheses of Theorem 10.4. A family of such arms has the stated estimate when its full comparison arms satisfy the physical overlap bound. Disjoint source intervals alone do not imply that bound.*

*Proof.* For $\xi \in I$ and $\eta \in J$, monotonicity gives $a(\eta) \le a(\xi)$ and $b(\eta) \ge b(\xi) + r$. Every point of the translated open source segment is therefore strictly between the comparison ends. The capacity hypothesis is (10.8), with the indicator of $J$ as comparison weight. □

**Proposition 10.6** (An actually reached level-width criterion)**.** *Suppose an actual arm has*

$$a(x) = a_* - \eta\psi(x), \qquad b(x) = a_* + d + \psi(x), \qquad 0 \le \eta \le 1, \quad 0 \le d \le \delta/2,$$

*where $\psi$ is strictly increasing and reaches $\delta/2, 2\delta, 3\delta$ at $x_1, x_2, x_3$ before a true boundary. Assume $\psi$ is continuously differentiable on $[x_2, x_3]$ and $x\psi'(x)/\psi(x) \le K$ there, with $K > 0$. Suppose, for some $Q, L, l > 0$, that $c^\varepsilon \le QL$ on $I = (0, x_1)$ and $p_\vee \ge Ql$ on $J = (x_2, x_3)$. Then the preceding corollary applies with*

$$r = \delta, \qquad \Delta = 8\delta, \qquad \kappa = \frac{LK}{l \log(3/2)}. \tag{10.12}$$

*Proof.* The outgoing increment from $x_1$ to $x_2$ is $3\delta/2$. On $J$ the residence is $d + (1+\eta)\psi(x) < 13\delta/2 < 8\delta$. Moreover

$$\log(3/2) = \int_{x_2}^{x_3} \frac{\psi'(x)}{\psi(x)}\, dx \le K \log(x_3/x_2) \le K(x_3 - x_2)/x_1.$$

Hence $\int_I c^\varepsilon \le QLx_1 \le \kappa Ql(x_3 - x_2) \le \kappa \int_J p_\vee$, as required. □

*Remark* 10.7 (Scope and rates). For $\eta > 0$ both roofs in Proposition 10.6 move. Taking $\psi(x) = x^k$ verifies its geometry with $K = k$, including zero endpoint velocities at $x = 0$ when $k > 1$. This is a diagnostic model of the comparison, not an asserted new Lorentz realization or a uniform bound on the orders of all arms. The theorem on actual arms uses exactly the stated reach, trace and overlap hypotheses. It does not extend a segment past its physical boundary.

Under (2.5), $\Delta \le 8\delta$ gives $\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta = O(\varepsilon^{1/16})$. Thus bounded $N\kappa$ gives $O(B^{-1/192})$; the larger budget $N\kappa = O(\varepsilon^{-1/32})$ gives $O(B^{-1/384})$. The width $r$ may be arbitrarily small at a fixed outer stage but must be fixed before the collision limit and uniform over its family.

## 11. Unallocated capacity in the original microscopic source

A comparison family need not satisfy full envelope balance on every block. The correct decomposition then retains the uncovered part as a positive source, rather than discarding the block or replacing its trace by the comparison envelope.

**Theorem 11.1** (Fractional comparison capacity)**.** *Assume Definition 10.2, without* (10.8)*. Put*

$$\beta_j = \begin{cases} \min\{1, \kappa P_j / C_j\}, & C_j > 0, \\ 0, & C_j = 0. \end{cases} \tag{11.1}$$

*Then* $\sum_j \beta_j T_j$ *satisfies* (10.9)–(10.10)*. At every finite count,*

$$\sum_j T_j = \sum_j \beta_j T_j + \sum_j (1-\beta_j) T_j, \qquad (1-\beta_j) C_j = (C_j - \kappa P_j)_+. \tag{11.2}$$

*For a fixed comparison family, these are the largest scalar fractions in* $[0,1]$ *for which the envelope-balance certificate holds block by block.*

*Proof.* Replace $s_j$ by $\beta_j s_j$ in Definition 10.2. Its new source capacity is $\beta_j C_j \le \kappa P_j$; no comparison weight or physical segment has changed. Theorem 10.4 applies. The displayed identities follow from (11.1). If a scalar fraction $\gamma_j \in [0,1]$ satisfies $\gamma_j C_j \le \kappa P_j$ and $C_j > 0$, then $\gamma_j \le \beta_j$. This is optimality for the stated capacity certificate, not an optimality claim over all possible physical comparison constructions. □

**Corollary 11.2** (Two endpoints improve one-ended capacity)**.** *For any fixed family, replacing* $P_j$ *by either* $P_j^- := \int w_j f_-\, d\theta$ *or* $P_j^+ := \int w_j f_+\, d\theta$ *cannot increase* $\beta_j$*. Consequently use of the two-ended envelope cannot enlarge the unallocated source for that family and budget.*

*Proof.* Pointwise $p_\vee \ge f_-$ and $p_\vee \ge f_+$. Integrate and use monotonicity of the function $x \mapsto \min\{1, \kappa x / C_j\}$ when $C_j > 0$. Zero-capacity blocks contribute zero under either choice. □

We apply the result to the original physical remainder, not to an auxiliary source. Fix the outer parameter $B$ and a previously fixed $0 < h_B < \delta(B)$. Let $\widetilde{H}_B$ denote the original unmatched Borel selector of Theorem 8.1: it retains the failure source outside both older recovered families, with their first-membership convention, and is restricted to $0 < \ell < h_B$. In particular

$$\mathsf{U}_B^{\mathrm{micro}}(t) = \int c^{\varepsilon(B)}(\theta) \mathbf{1}_{\{a(\theta) < t < b(\theta)\}} \widetilde{H}_B(\Psi(\theta, t))\, d\theta.$$

All evaluations of this selector remain at the original target. Choose disjoint angular source sets $I_{B,j}$ and comparison weights satisfying Definition 10.2 with $s_j = \mathbf{1}_{I_{B,j}}$. Put $T_{B,j} = T_j$ with target selector $\widetilde{H}_B$, and define

$$\mathsf{V}_B(t) = \sum_j \beta_{B,j} T_{B,j}(t), \tag{11.3}$$

$$\mathsf{R}_B(t) = \int c^{\varepsilon(B)}(\theta)\mathbf{1}_{\{a<t<b\}}\widetilde{H}_B(\Psi(\theta,t))\left(1 - \sum_j \beta_{B,j}\mathbf{1}_{I_{B,j}}(\theta)\right) d\theta. \tag{11.4}$$

The expression in parentheses belongs to $[0,1]$. It equals 1 outside the selected angular source sets and equals $1-\beta_{B,j}$ on the $j$th such set.

**Theorem 11.3** (Exact residual and original pointwise reduction)**.** *The original finite-count source has the positive decomposition*

$$\mathsf{U}_B^{\mathrm{micro}} = \mathsf{V}_B + \mathsf{R}_B. \tag{11.5}$$

*Suppose the older costs in Theorem 8.1 obey its stated budget. Suppose also that the new comparison family has $0 < r_B < \Delta_B \le 8\delta(B)$ and $N_B\kappa_B \le C\varepsilon(B)^{-1/32}$, uniformly in the collision count, radius and exact labels. Each of these parameters, including $h_B$, is fixed before the collision limit. Then*

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B) + C_M B^{-1/384}. \tag{11.6}$$

*Along this schedule the unchanged raw pointwise assertion* (1.5) *is equivalent to*

$$\mathcal{H}_M(\mathsf{R}_B) \longrightarrow 0 \quad \textit{for every fixed } M. \tag{11.7}$$

*Proof.* The integral definition and disjointness of the source sets prove (11.5); for countably many sets use Tonelli. Theorem 11.1 and Remark 10.7 give $\mathcal{H}_M(\mathsf{V}_B) \le C_M B^{-1/384}$. The preceding exact scalar reduction and subadditivity of the nonnegative essential-height functional therefore imply (11.6). If (11.7) holds, this bound proves the original assertion. Conversely that assertion implies $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) \to 0$ by Theorem 8.1; the pointwise order $0 \le \mathsf{R}_B \le \mathsf{U}_B^{\mathrm{micro}}$ gives necessity. □

The two-ended estimate pays the entire difference between the comparison envelope and its original positive paired trace, even when the latter is zero. It does not claim that every physical source has available comparison capacity. Formula (11.4) retains all unmatched sheets, including sheets without a usable actual corridor, and the full unallocated fraction on every capacity-deficient block. In particular no inverse-residence or logarithmic moment is silently inserted into the proof. The positive limit (11.7) is the remaining estimate for the organizing raw pointwise theorem; it has not been established by the comparisons proved in this article.

*Remark* 11.4 (Compatibility with the earlier insertion criteria)*.* The untruncated inverse-residence criterion and the logarithmic criterion retained above remain sufficient tests on this smaller positive remainder whenever their respective hypotheses hold. Indeed multiplication by the nonnegative residual fraction in (11.4) preserves pointwise domination by the original weighted source. Neither finiteness of the weighted source nor its uniform height is inferred merely from this domination.

## 12. Canonical comparison capacity at the original roof

The source and comparison trajectories need not be matched on an entire angular block. We instead compare them at the roof at which the source is evaluated. This distinction permits a canonical construction, with physical overlap one, on the full comparison space supplied by the inherited regular-segment hypotheses.

Fix the collision count, radius, exact labels and $\varepsilon > 0$. All integrals below are on the original disjoint physical pairing space $\Theta_{\mathrm{phys}}$. We use its Borel segment endpoints and the finite traces in (10.1), together with Lemma 10.1. These are precisely the inherited regular-segment inputs, not new assertions about unverified singular extensions. If those inputs have been verified only on a specified Borel comparison subspace, every comparison integral in this section is restricted to that subspace; the source outside it is retained. Write $\ell = b - a$ and fix $0 < r < \Delta$ before taking a collision limit.

**Definition 12.1** (The complete admissible pool)**.** At the original roof $t$, put

$$\begin{aligned} q_+(t,\theta) &= \mathbf{1}_{\{0<\ell(\theta)<\Delta,\ a(\theta)<t,\ t+r<b(\theta)\}},\\ q_-(t,\theta) &= (1-q_+(t,\theta))\mathbf{1}_{\{0<\ell(\theta)<\Delta,\ a(\theta)<t-r,\ t<b(\theta)\}},\\ Q_r(t,\theta) &= q_+(t,\theta) + q_-(t,\theta). \end{aligned} \tag{12.1}$$

Thus a sheet is counted once even when both orientations are available. Define

$$P_r(t) = \int Q_r p_\vee \, d\theta, \qquad D_r(t) = \int Q_r c^\varepsilon \, d\theta, \qquad E_r(t) = \int Q_r e_\partial \, d\theta. \tag{12.2}$$

In particular $P_r = D_r + E_r$. The pool contains every admissible physical segment containing one of the two closed roof windows $[t, t+r]$ and $[t-r, t]$ in its interior. It is not a selected collection of angular source blocks.

**Lemma 12.2** (Measurability and one-copy localization)**.** *The functions in* (12.1) *are jointly Borel and satisfy* $0 \le Q_r \le 1$. *Their integrals have common measurable roof versions at the fixed collision count, radius and exact labels. On* $\{q_+ > 0\}$ *the segment contains* $[t, t+r]$*; on* $\{q_- > 0\}$ *it contains* $[t-r, t]$. *On* $\{Q_r > 0\}$ *its entire interior lies in* $(t-\Delta, t+\Delta)$.

*Proof.* The indicator conditions are Borel inequalities between the inherited endpoint functions and $t$. The factor $1 - q_+$ disjointizes the two orientations. Integration of a nonnegative jointly Borel function against the fixed sigma-finite angular measure gives a measurable function of $t$. The asserted window inclusions follow from the strict inequalities. Finally $a < t < b$ and $b - a < \Delta$ imply $t - \Delta < a < b < t + \Delta$. All statements concern the original physical measure, so a disjoint refinement of its charts changes none of the integrals. We use the inherited common coarea versions; a null set uniform over all radii is neither needed nor asserted. □

**Theorem 12.3** (Canonical finite-count envelope bound)**.** *At finite count, on the original roof versions,*

$$P_r(t) \le \frac{1}{r}\int_{t-r}^{t+r} \mathsf{T}^\varepsilon_\Delta(u)\, du + C(1+\varepsilon^{-1}) \int_{t-\Delta}^{t+\Delta} v^0(u)\, du. \tag{12.3}$$

*Consequently, for any previously fixed* $K > 0$,

$$\mathcal{H}_M(KP_r) \le CK\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{12.4}$$

*No source-block balance, angular continuation, monotonicity of either endpoint roof, or independently supplied overlap bound is required.*

*Proof.* For the positive orientation, Tonelli's theorem gives

$$r\int q_+(t,\theta)c^\varepsilon(\theta)\, d\theta = \int_t^{t+r}\int q_+(t,\theta)c^\varepsilon(\theta)\mathbf{1}_{\{a(\theta)<u<b(\theta)\}}\, d\theta\, du.$$

The analogous identity holds on $(t-r, t)$ for $q_-$. Dropping the comparison weights on their respective half-windows proves

$$D_r(t) \le \frac{1}{r}\int_{t-r}^{t+r} \mathsf{T}^\varepsilon_\Delta(u)\, du.$$

The weights depend on $t$, but $t$ is fixed throughout these integrals; they are never evaluated at $u$ or differentiated. Lemma 10.1 and Lemma 12.2 give

$$\begin{aligned} E_r(t) &\le C(1+\varepsilon^{-1}) \int Q_r(t,\theta) \int_{a(\theta)}^{b(\theta)} A(u,\theta) I_m(u,\theta)/z(u,\theta)\, du\, d\theta \\ &\le C(1+\varepsilon^{-1}) \int_{t-\Delta}^{t+\Delta} v^0(u)\, du. \end{aligned}$$

The last step uses the original one-copy physical summation and $Q_r \le 1$, not disjointness of auxiliary angular blocks. Adding the two inequalities proves (12.3). All sums are nonnegative and are completed before any norm or collision limit. The inherited fixed-count local integrability makes the displayed quantities finite for almost every roof. Finally use (2.1) at $r$ and (2.2) at $\Delta$. The factors are respectively $2K$ and $2CK(1+\varepsilon^{-1})\Delta$. This proves (12.4) in the prescribed order of limits. □

Let $0 \le H \le 1$ be an original Borel selector and put

$$U(t) = \int c^{\varepsilon}(\theta) \mathbf{1}_{\{a(\theta)<t<b(\theta)\}} H(\Psi(\theta,t))\, d\theta.$$

Assume the inherited local finiteness of this positive source. Set

$$\begin{aligned} &V_{r,K}(t) = \min\{U(t), KP_r(t)\}, \qquad && R_{r,K}(t) = (U(t) - KP_r(t))_+, \\ &\beta_{r,K}(t) = \begin{cases} \min\{1, KP_r(t)/U(t)\}, & U(t) > 0, \\ 0, & U(t) = 0. \end{cases} \end{aligned} \tag{12.5}$$

Definitions may be completed arbitrarily on the common roof-null set where finiteness fails.

**Corollary 12.4** (An exact original-source allocation)**.** *The multiplier $\beta_{r,K}$ is measurable and takes values in $[0,1]$. At finite count,*

$$U = V_{r,K} + R_{r,K}, \qquad V_{r,K}(t) = \int c^{\varepsilon} \mathbf{1}_{\{a<t<b\}} H(\Psi(\theta,t)) \beta_{r,K}(t)\, d\theta. \tag{12.6}$$

*Replacing $\beta_{r,K}$ by $1-\beta_{r,K}$ gives the corresponding integral formula for $R_{r,K}$. Moreover $V_{r,K}$ satisfies the right sides of* (12.3) *and* (12.4)*, multiplied by $K$ in the former.*

*Proof.* Measurability follows from Lemma 12.2. The identities follow pointwise from the minimum and positive-part formulas. Since $V_{r,K} \le KP_r$, Theorem 12.3 applies. The endpoint maximum supplies only comparison capacity. The measure on the source side still has its original minimum trace and its original target selector; neither has been moved to another roof. □

## 13. Maximal roofwise capacity and the microscopic residual

The preceding pool has a precise maximality property. We specify the class of certificates, since no optimality over all conceivable physical transports is asserted.

**Definition 13.1** (A roofwise window certificate)**.** Fix the same $r, \Delta, K$ and the same admissible comparison space as in Definition 12.1. A certificate for an accepted source height $Z(t)$ consists of nonnegative measurable weights $w_+(t,\theta), w_-(t,\theta)$ with $w_+ + w_- \le 1$. The positive weight is supported on segments of length less than $\Delta$ containing $[t, t+r]$ in their interior, and the negative weight on those containing $[t-r, t]$. Require

$$0 \le Z(t) \le U(t), \qquad Z(t) \le K \int (w_+ + w_-) p_{\vee}\, d\theta. \tag{13.1}$$

Weights may depend measurably on the original roof. No selector is transported, no derivative of a selector is used, and the comparison side has the same radius, collision count and exact labels.

**Theorem 13.2** (Attained maximality)**.** *Among all certificates in Definition 13.1, the largest accepted height, pointwise almost everywhere, is $V_{r,K}$. It has an explicit measurable certificate. Thus $R_{r,K}$ is the smallest residual in this specified class.*

*Proof.* Admissible supports and the one-copy constraint imply $w_+ + w_- \le Q_r$. Therefore (13.1) gives $Z \le \min\{U, KP_r\} = V_{r,K}$. For attainment, on $P_r > 0$ set $\gamma(t) = V_{r,K}(t)/(KP_r(t))$, and set $\gamma = 0$ on $P_r = 0$. Use $w_+ = \gamma q_+$ and $w_- = \gamma q_-$. These are measurable, their sum is at most one, and their certified capacity is exactly $V_{r,K}$. When $P_r = 0$ both sides vanish. The source allocation is the distinct multiplier $\beta_{r,K}$ in (12.6). This proves attainment without a selection theorem or an optimization over an uncountable list of angular families. □

**Theorem 13.3** (Domination of fixed-family allocations)**.** *Let a revision-107 family act on the same source $U$, with comparison lengths below $\Delta$, window width at least $r$, overlap at most $N$, and block budget $\kappa$. Its accepted source $V^{\mathrm{old}}$ then satisfies*

$$V^{\mathrm{old}}(t) \le V_{r,N\kappa}(t), \qquad R_{r,N\kappa}(t) \le U(t) - V^{\mathrm{old}}(t). \tag{13.2}$$

*The statement holds for every such family; no family maximizing the old blockwise criterion has to exist or be selected.*

*Proof.* At a roof $t$, retain only blocks with $T_j(t) > 0$. Lemma 10.3 puts their comparison supports inside $\{Q_r(t,\cdot) = 1\}$. The blockwise fraction obeys $\beta_j T_j(t) \le \beta_j C_j \le \kappa P_j$. For a finite collection of active blocks,

$$\sum_j \beta_j T_j(t) \le \kappa \int p_\vee(\theta) \sum_j w_j(\theta)\, d\theta \le N\kappa P_r(t).$$

The full inequality follows by monotone convergence and the inherited common versions for the countable family. Since source sets are disjoint and $0 \le \beta_j \le 1$, also $V^{\mathrm{old}} \le U$. Taking the minimum proves the first inequality; subtracting from $U$ proves the second. This is domination at the original roof, not merely an inequality between total angular capacities. □

**Proposition 13.4** (The new certificate is strictly broader)**.** *There are regular interval-source data for which no nonzero whole-source-window block allocation of width $r$ is possible but the roofwise certificate accepts the entire selected source on a roof interval of positive length.*

*Proof.* Take $r = 1/10$, $s = 1/1000$, and two disjoint angular sets of measure one, each with constant minimum and maximum trace equal to one. On the first set the physical interval is $(0, s)$; on the second it is $(3s/8, r + 7s/8)$. Select only the first set and take $\Delta > r + s/2$, $K = 1$. For $3s/8 < t < 7s/8$ the second interval contains $[t, t + r]$ in its interior. Hence $U(t) = 1$, $P_r(t) \ge 1$ and $V_{r,1}(t) = 1$ on that roof interval. Neither available interval has length at least $r + s$. Consequently neither can contain a translate by $[0, r]$ or $[-r, 0]$ of the entire first interval. Subdividing its angular set does not alter its interval. Thus no nonzero revision-107 block comparison is available for this selected source at width $r$. All traces are constant, so the endpoint variation is zero. These data demonstrate strictness of the certificate classes. They are not asserted to constitute a new Lorentz realization or an asymptotic example for the original collision process. □

Return now to the original unmatched source $U = \mathsf{U}_B^{\mathrm{micro}}$ of Theorem 8.1. Its selector $\widetilde{H}_B$ and its restriction $0 < \ell < h_B$ are unchanged. Set

$$\Delta_B = 8\delta(B), \qquad 0 < r_B < \Delta_B, \qquad 1 \le K_B \le C\varepsilon(B)^{-1/32}. \tag{13.3}$$

Here $r_B$, $h_B$ and $K_B$ are fixed at each outer stage before the collision limit, uniformly over the radius and exact labels. Define the canonical residual on the original source by

$$\begin{aligned}\mathsf{R}_B^{\mathrm{can}}(t) &= \big(\mathsf{U}_B^{\mathrm{micro}}(t) - K_B P_{r_B}(t)\big)_+ \\ &= \int c^{\varepsilon(B)}(\theta)\mathbf{1}_{\{a(\theta)<t<b(\theta)\}}\widetilde{H}_B(\Psi(\theta,t))(1-\beta_{r_B,K_B}(t))\,d\theta.\end{aligned} \tag{13.4}$$

The residual does not depend on a choice of angular source blocks. It retains every original positive weight not paid by the complete admissible roofwise capacity pool.

**Theorem 13.5** (Canonical reduction of the unchanged assertion)**.** *Under the inherited scalar and physical-source inputs, and the older budgets of Theorem 8.1, the construction* (13.3) *satisfies*

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{can}}) + C_M B^{-1/384}. \tag{13.5}$$

*The original assertion* (1.5) *is equivalent to*

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{can}}) \longrightarrow 0 \quad \textit{for every fixed } M. \tag{13.6}$$

*At the same or a smaller fixed window and the same total cost,* $\mathsf{R}_B^{\mathrm{can}}$ *is pointwise no larger than any revision-107 residual covered by Theorem 13.3.*

*Proof.* Corollary 12.4 decomposes the complete finite-count microscopic source into its accepted and residual parts. The inherited schedule has $\delta = O(\varepsilon^4)$. Therefore Theorem 12.3 gives

$$\mathcal{H}_M(V_{r_B,K_B}) \le C\varepsilon^{-1/32}\{\varepsilon^{1/16} + O(\varepsilon^3)\} = O(\varepsilon^{1/32}) = O(B^{-1/384}).$$

The inherited scalar reduction and subadditivity of $\mathcal{H}_M$ prove (13.5). The displayed decay implies the original assertion by sending $B$ to infinity. Conversely the original assertion implies $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) \to 0$ by Theorem 8.1; use $0 \le \mathsf{R}_B^{\mathrm{can}} \le \mathsf{U}_B^{\mathrm{micro}}$. The pointwise comparison is Theorem 13.3. □

**Proposition 13.6** (Finite-count exhaustion and its quantifiers)**.** *Fix the collision count, radius, exact labels and* $\Delta$*. Suppose the selected source is supported on the admissible comparison space with* $0 < \ell < \Delta$*, and* $K \ge 1$*. Then* $R_{r,K}(t)$ *decreases to zero for almost every* $t$ *as* $r \downarrow 0$*. This conclusion alone does not establish* (13.6)*.*

*Proof.* For every interior point of a positive-length segment, a sufficiently small one-sided window lies in that segment. Hence, as $r \downarrow 0$, $Q_r(t,\theta)$ increases to $\mathbf{1}_{\{0<\ell(\theta)<\Delta,\ a(\theta)<t<b(\theta)\}}$. Monotone convergence and $p_\vee \ge c^\varepsilon$ give $\lim_{r\downarrow 0} P_r(t) \ge U(t)$. The positive-part formula proves the asserted finite-count limit. Here the collision count was fixed first. In (13.6), $r_B$ must be fixed first, followed by the complete physical sum, the prescribed essential suprema and collision limsup, with $B$ outermost. No interchange of these two orders has been proved or used. □

The canonical construction removes the need to choose angular blocks and to prove a separate overlap bound for their comparison supports. It pays the complete endpoint trace loss and exhausts the available capacity within the stated window-certificate class. It does not prove that this capacity has a collision-uniform surplus on the original microscopic source. In particular (13.6) remains a substantive estimate on that source, not a consequence of finite-count exhaustion. Nor does the construction verify additional singular trace extensions outside the inherited physical-source inputs. The original raw pointwise assertion, rather than a reformulated comparison problem, remains the organizing assertion of the paper.

## Appendix A. Dependencies and quantifiers of the canonical construction

Theorem 12.3 uses only the inherited Borel physical pairing space, finite one-sided endpoint traces, the regular-segment variation bound, the one-copy uncut volume, and the fixed-window paired and uncut average estimates. Its construction supplies the comparison weights explicitly; overlap one is an identity, not a new geometric assumption. Theorem 13.2 concerns exactly Definition 13.1, at fixed $r, \Delta, K$. It makes no claim of optimality over arbitrary comparison geometries, transport costs, or changing windows. Theorem 13.3 is a pointwise theorem, so it does not confuse optimization of total angular mass with optimization in the required roof topology.

All source multipliers are evaluated at the original target roof. At each fixed collision count, radius and exact labels, integration and the positive physical sum precede every essential supremum. The outer stage fixes $\varepsilon(B), h_B, r_B, \Delta_B, K_B$. After those choices come the radius and exact-label suprema and the collision limsup; $B \to \infty$ is last. Finite-count exhaustion in Proposition 13.6 is not substituted for this ordered estimate. No inverse-residence or logarithmic moment is claimed for the Lorentz source by the new construction.

The original singular trace extensions still require their stated physical hypotheses. An endpoint is not admitted to the comparison space merely because it has been given a singular or stationary name. Any source not paid by admissible capacity remains, with its original positive weight, in (13.4). Thus the construction addresses comparison selection and overlap, but the collision-uniform shortage estimate (13.6) requires a further proof on the original Lorentz family.

## Appendix B. Proof dependencies and the complete remaining source

The original event and target are (1.1) and (1.5). The inherited inputs include the complete positive paired measure, its fixed-width concentration, actual uncut optical volume, weighted variation, regular critical trace and the original scalar schedule. Their historical proofs remain in the archive; no new spectral or pressure theorem is assumed.

Endpoint inclusion and integrated positive-trace balance give Theorem 3.1 at finite collision count. The critical incoming normalization and the width of one actual comparison interval give Lemma 4.1. If the outgoing trace is too small, Theorem 4.3 pays its entire shortage by the original uncut volume in one common roof window. The resulting height estimate needs no outgoing trace-fraction lower bound. Corollary 5.2 then removes the entire selected critical-arm contribution from the old three-clock failure. Physical overlap is counted on the disjoint angular sheet space, not by counting words or strata.

The finite-order condition is local to an actual arm. Its comparison levels must be reached before any true boundary, and its geometric ratio must meet the stated uniform or outer-budget bound. No uniform order, reach or complete physical coverage is inferred from finite-count definability. The stationary model checks the distinction from the old clocks; it is not an additional Lorentz realization.

The complete remaining source is positive and lies outside the recovered physical families. Vanishing of that source in the original essential-height norm is still required for the organizing theorem. The complete preceding article and all earlier mathematical cores are reproduced in the supplement, with the two revision-88 variants separately identified and preserved.

## Appendix C. Dependencies, physical coverage and the remaining exact source

The original event and raw exact-label pointwise target remain (1.1) and (1.5). The complete positive paired coarea source and uniform fixed-window input are (3.1) and (2.1); the complete uncut optical volume and weighted one-sided trace variation are (2.2) and (2.3).

| **Original positive source** | **Proved payment** | **Residual** |
|---|---|---|
| Certified relative/endpoint-distance clocks | Inherited v103 positive recovery | Uncertified clock failures |
| Full actual v104 critical-arm reach | Corollary 5.2 | Unreached arms |
| Adaptive smaller reached critical arms | Theorem 7.1 | Arms not reaching $3\rho/4$ |
| Every flat, singular, reach-deficient or nonstationary sheet with $\ell \geq h_B$ | Theorem 6.1 | None at this fixed resolution |
| Original unmatched sheets with $0 < \ell < h_B$ | Exact source (8.1) | Positive microscopic height not proved |

The moment insertion in (8.3) is a sufficient new source-level target, not an already proved Lorentz estimate. The full preceding v104 article, all 252 earlier mathematical cores, complete historical derivation and both separate v88 variants remain unchanged in the inherited archive.

## Appendix D. Coverage and order of limits

The organizing event and pointwise target are still (1.1) and (1.5). The following account separates established estimates for positive source classes from the remaining estimate on their complement.

Positive residence. The persistence theorem already estimates every selected physical sheet with $\ell \geq h_B$ at a previously fixed $h_B > 0$. This includes flat isolated pulses, singular endpoint types and reach-deficient sheets whenever their total residence is at least $h_B$. Uniformity of the leading constant in a fixed $h_B$ does not permit replacing $h_B$ by a collision-dependent number.

Stationary and moving comparison families. The older critical-arm and adaptive-reach theorems retain their proved classes. Theorem 10.4 additionally applies to nonstationary entrances, nonstationary exits and weighted comparison overlaps, provided actual interval inclusion and comparison capacity hold. Its proof uses only regular-segment variation and the original volume source. A singular endpoint type is therefore not excluded by its name, but it must satisfy those actual input and comparison hypotheses; the regular critical Hessian formula is not extended to singular endpoints by terminology.

Trace loss versus capacity shortage. For each comparison block the trace loss is $E_j = P_j - D_j$ and is paid in full by (10.9). The remaining capacity shortage is $(C_j - \kappa P_j)_+$, not $(C_j - \kappa D_j)_+$. Theorem 11.1 proves this distinction even when $D_j = 0$. Formula (11.4) retains both the uncovered microscopic sheets and the unallocated fractions. It is not a signed error, a moment-truncated substitute or an omitted face term.

Actual reach and uniformity. All comparison intervals lie on actual sheets of the same exact event. The overlap bound is imposed on their full incident-sheet supports; disjoint starting sets alone do not establish it. Proposition 10.6 is a sufficient verification on a stated class of arms. Finite-count definability alone supplies neither its level-width constant nor a budget uniform in $m, R, n, k$. Consequently the proposition is not a classification of the entire physical source.

Quantifiers. First fix $M, B, \varepsilon(B), h_B, r_B, \Delta_B$ and all other auxiliary bands. At every finite $m, R, n, k$ form the entire physical sum and its exact nonnegative decomposition. Next take the roof essential supremum, the supremum in radius and exact labels, and the collision limit superior in the order defining $\mathcal{H}_M$. Only afterwards may the fixed auxiliary bands be enlarged or a previously fixed residence threshold be decreased. The outer parameter $B$ tends to infinity last. The estimate for a complete shrinking-residence tail is not obtained by exchanging the collision limit with $h \downarrow 0$. The conditional-law conclusions retain their original scalar-first order.

Historical provenance and present conclusion. The submission-state review of revision 103 predates its final mathematical push; it is not a substantive rejection of that later source. The substantive report addressed here is the revision-104 report. Revisions 105 and 106 are retained as subsequent author work, not described as externally certified mathematics. Every inherited numbered mathematical module is retained. The malformed paid-arm reference in the active introduction has been corrected; the historical source snapshot remains byte-preserved.

The new conclusions are Theorems 10.4, 11.1 and 11.3, with their stated hypotheses. They directly enlarge the paid positive source and sharpen its exact remainder. The global estimate (11.7) is not proved here. In particular the raw pointwise theorem, unrestricted conditioned laws and an independent specialist verification are not certified by the finite checks or by successful typesetting.

## References

[1] L. Ambrosio, N. Fusco and D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems*, Oxford Mathematical Monographs, Clarendon Press, Oxford, 2000.

[2] R. Cluckers and D. J. Miller, *Stability under integration of sums of products of real globally subanalytic functions and their logarithms*, Duke Math. J. **156** (2011), no. 2, 311–348; arXiv:0911.4373v1, Theorem 1.3.

[3] M. Coste, *An introduction to o-minimal geometry*, lecture notes, November 1999; Dottorato di Ricerca in Matematica, Università di Pisa, Istituti Editoriali e Poligrafici Internazionali, 2000.

[4] V. Baladi and M. F. Demers, *Thermodynamic formalism for dispersing billiards*, J. Mod. Dyn. **18** (2022), 415–493; arXiv:2009.10936v3, Section 1.1, equations (2.4), (2.11), and the proof of Lemma 3.8.

[5] M. F. Demers and H.-K. Zhang, *A functional analytic approach to perturbations of the Lorentz gas*, arXiv:1210.1261v1 (2012), Theorem 2.5, Remark 2.7 and Lemma 3.2(b); doi:10.1007/s00220-013-1820-0.